Recommending UCLA Study Spots

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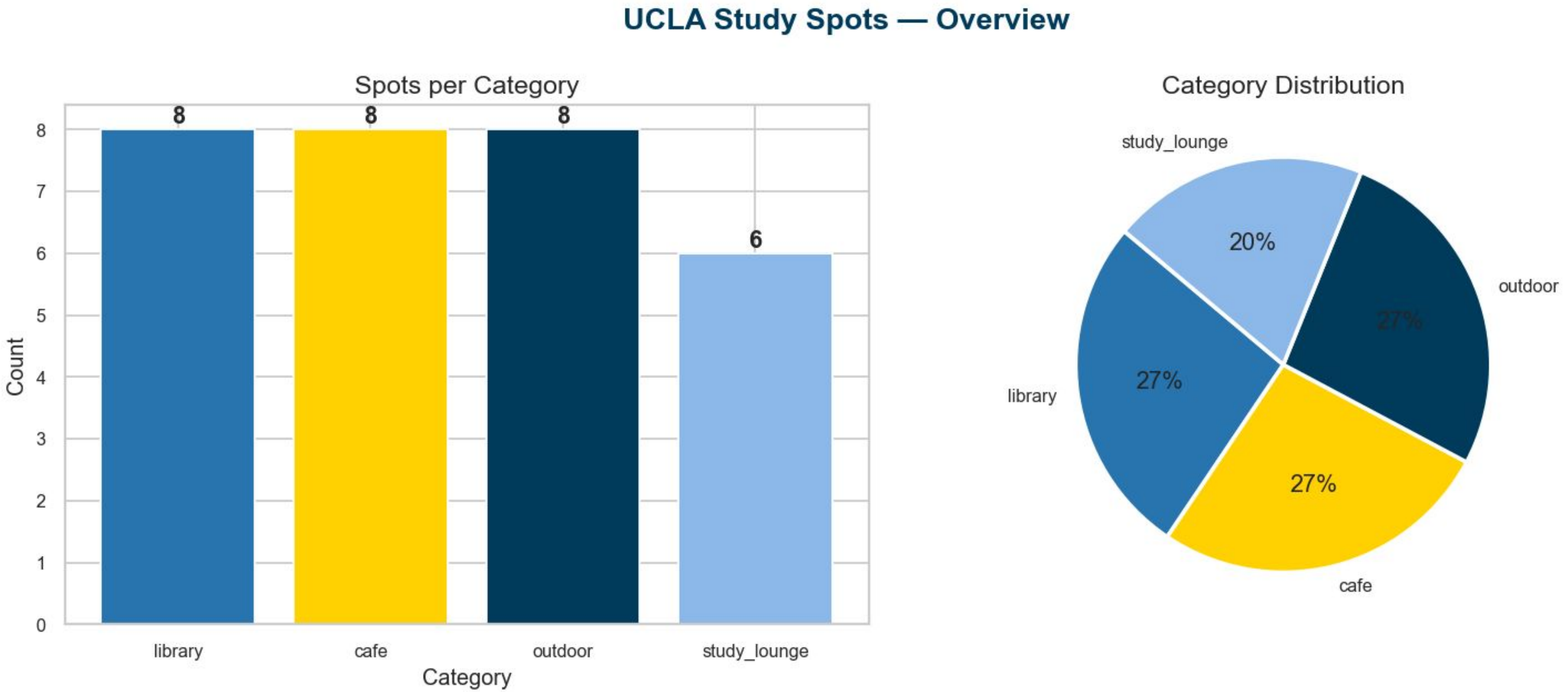
Project Overview and Motivation

- UCLA Study Spots Recommendation System
  - Recommends cafes, study lounges, libraries, or outdoor areas to user
- Problem:
  - May only go to one study spot, but it won't always be available
  - Wants a study spot that is moderately quiet, but doesn't know which to choose
- Motivation:
  - During midterm and final exam season, many study spots may be filled
  - Some may want a change of environment to help studying
  - Makes it easier to choose a place to study
- Target users: UCLA students

Data Collection Approach and Progress

- Data sources: library.ucla.edu, asucla.ucla.edu/locations, housing.ucla.edu
- Collection Method: Web scraping these websites
- Current Progress: 30 different study spots on campus
- Data features:
  - Name
  - Category (library, study_lounge, cafe, outdoor)
  - Noise_level
  - Features (wifi, 24_hours,...)

Exploratory Data Analysis



Proposed Modeling Approach

- Model: KNN, Random Forest
- Target: name
- Inputs: category, address, description, noise_level, wifi, indoor_outdoor
- Will measure success based on how much the recommendations match what the user wants (accuracy)
- Baseline approach:
    - Perform more EDA and test different models

Timeline and Next Steps

- Week 6: Complete data collection and EDA
- Week 7: Finish feature engineering and choose models
- Week 8: Train and evaluate models
- Week 9: Build Streamlit app
- Week 10: Final testing, deploy app

Challenges and Strategies:

- No previous user history -> use KNN to find spots that match desired features
- Time of day and context -> use random forest to filter and rank spots